

Name	Type	When It Works	Example Use
<code>mouseX</code>	Variable	Always	Current mouse X position
<code>mouseY</code>	Variable	Always	Current mouse Y position
<code>pmouseX</code>	Variable	Always	Previous mouse X position
<code>pmouseY</code>	Variable	Always	Previous mouse Y position
<code>mouseIsPressed</code>	Boolean Variable	While mouse button is held down	Draw while clicking
<code>mouseClicked()</code>	Event Function	Once after a click (press + release)	Button click
<code>mouseMoved()</code>	Event Function	When mouse moves without clicking	Track cursor
<code>mouseDragged()</code>	Event Function	When mouse moves while pressed	Paint program
<code>mousePressed()</code>	Event Function	When mouse button is pressed	Start an action
<code>mouseReleased()</code>	Event Function	When mouse button is released	Stop an action

1. `mouseX`

Stores the current X-coordinate of the mouse.

```
ellipse(mouseX, 200, 20, 20);
```

The circle moves left and right with the mouse.

2. `mouseY`

Stores the current Y-coordinate of the mouse.

```
ellipse(200, mouseY, 20, 20);
```

The circle moves up and down with the mouse.

3. **mouseX** and **mouseY** Together

```
ellipse(mouseX, mouseY, 20, 20);
```

The shape follows the mouse pointer.

4. **pmouseX**

Stores the previous X-position of the mouse.

```
line(pmouseX, pmouseY, mouseX, mouseY);
```

Useful for drawing smooth lines.

5. **pmouseY**

Stores the previous Y-position of the mouse.

```
line(pmouseX, pmouseY, mouseX, mouseY);
```

Combined with **pmouseX** to track movement.

6. **mouseIsPressed**

Returns **true** when the mouse button is held down.

```
draw = function() {  
  if (mouseIsPressed) {  
    ellipse(mouseX, mouseY, 20, 20);  
  }  
};
```

Used for drawing only when clicking.

7. mouseClicked()

Runs once when the mouse is clicked.

```
mouseClicked = function() {  
  println("Clicked!");  
};
```

Useful for buttons and games.

8. mouseMoved()

Runs whenever the mouse moves.

```
mouseMoved = function() {  
  println(mouseX + "," + mouseY);  
};
```

Tracks mouse movement.

9. mouseDragged()

Runs when the mouse moves while pressed.

```
mouseDragged = function() {  
  ellipse(mouseX, mouseY, 10, 10);  
};
```

Creates a paintbrush effect.

10. Distance from Mouse

```
var d = dist(mouseX, mouseY, 200, 200);
```

Calculates distance between mouse and a point.

Useful for hover effects and collision detection.

11. Object Following Mouse

```
ellipse(mouseX, mouseY, 50, 50);
```

Makes an object follow the cursor.

12. Change Color on Click

```
draw = function() {  
  if (mouseIsPressed) {  
    fill(255, 0, 0);  
  } else {  
    fill(0, 255, 0);  
  }  
  
  ellipse(200, 200, 100, 100);  
};
```

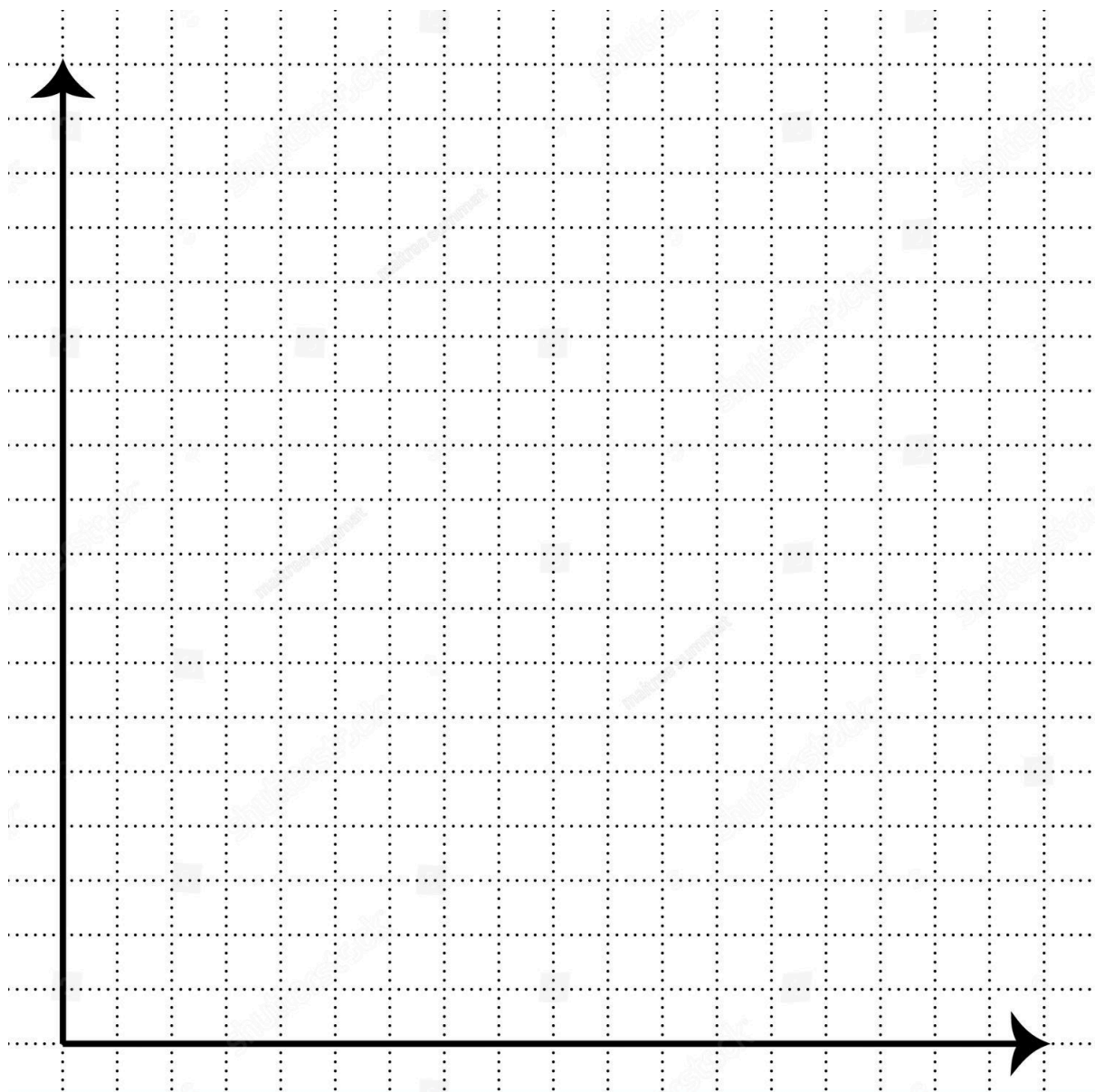
Changes color based on mouse state.

`ellipse(x, y, width, height)`

`fill(r,g,b)`

`mouseX, mouseY`

```
draw = function() {  
  ellipse(mouseX, mouseY, 10, 10);  
};
```



CONDITIONAL STATEMENT

If - statement

```
if (condition) {  
    // body or code  
}
```

```
var speed = 5;
```

```
If (speed == 5) {  
    Speed = speed + 10;  
}
```

>, <, =, >=, <=